



BMW

**Used Car Sales Analysis Dash
Board (2010-2024)**

AUTHOR AND TOOLS USED



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A BIS STUDENT AT AAST, INTERESTED IN DATA ANALYSIS AND UI/UX DESIGN. I STUDIED AT ROUTE ACADEMY AND GAINED HANDS-ON EXPERIENCE THROUGH INTERNSHIPS AT ROUTE, DHL, AND GASCOOL CO.



Kaggle

ABOUT KAGGLE

KAGGLE IS AN ONLINE PLATFORM OWNED BY GOOGLE THAT HOSTS DATASETS, COMPETITIONS, AND LEARNING RESOURCES FOR DATA SCIENCE AND MACHINE LEARNING. IT PROVIDES FREE ACCESS TO A WIDE VARIETY OF REAL-WORLD DATASETS, WHICH I USED AS THE SOURCE FOR THIS PROJECT.



MS Excel

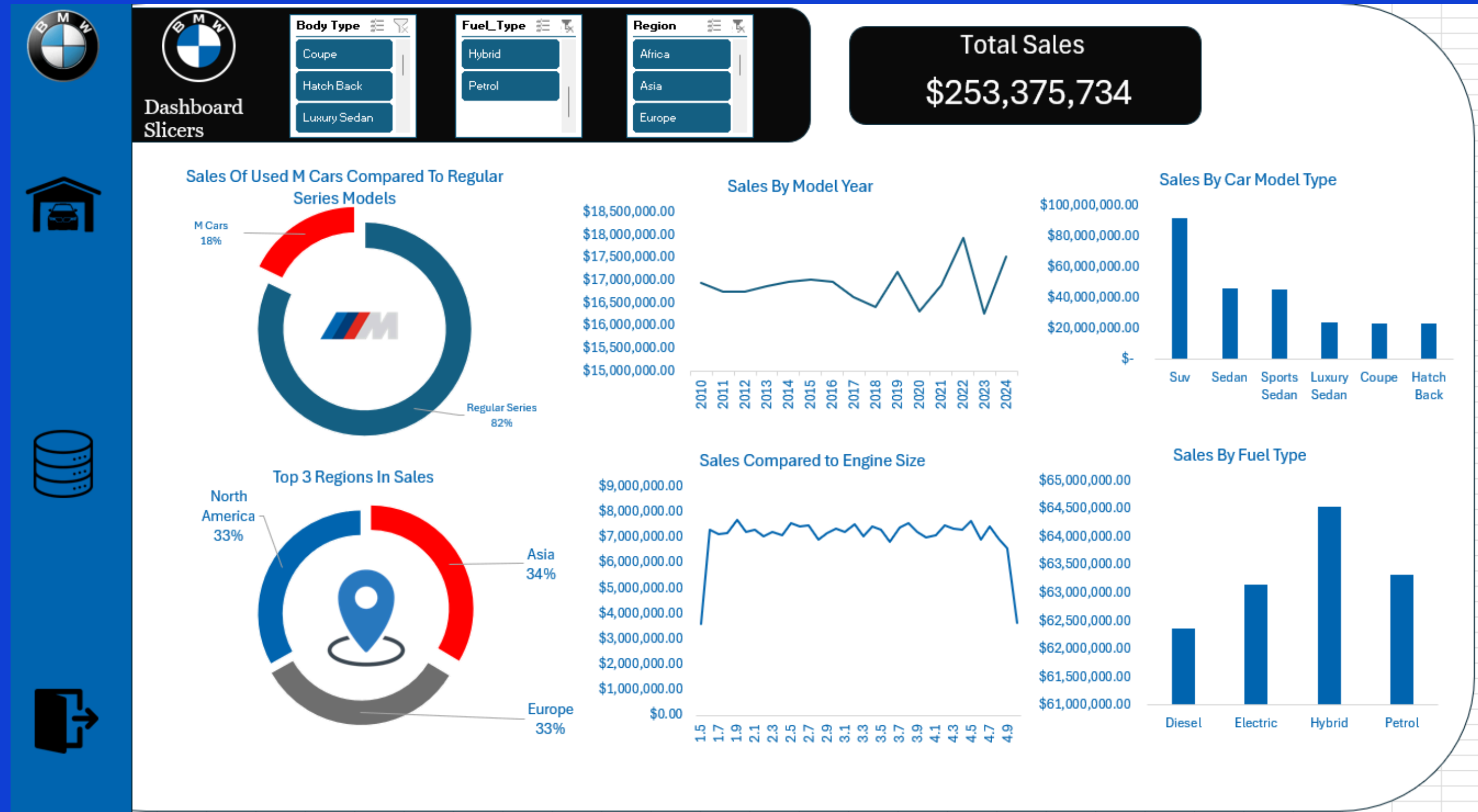
ABOUT MICROSOFT EXCEL

MICROSOFT EXCEL IS ONE OF THE MOST WIDELY USED TOOLS FOR DATA ORGANIZATION, ANALYSIS, AND VISUALIZATION. IN THIS PROJECT, I USED EXCEL TO CLEAN THE DATASET, PERFORM ANALYSIS, AND BUILD THE DASHBOARD.

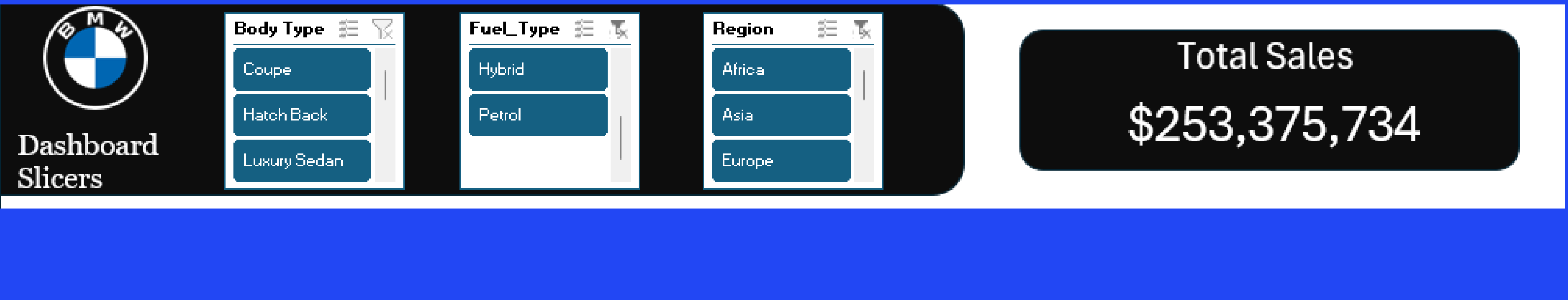
INTRODUCTION

This project analyzes BMW used car sales data from Kaggle. After cleaning and preparing the dataset in Excel, pivot tables, slicers, and visuals were used to explore sales by year, engine size, model type, fuel type, and region. The dashboard provides clear insights into BMW's used car sales performance and highlights key market trends.

DASHBOARD'S MAIN PAGE:



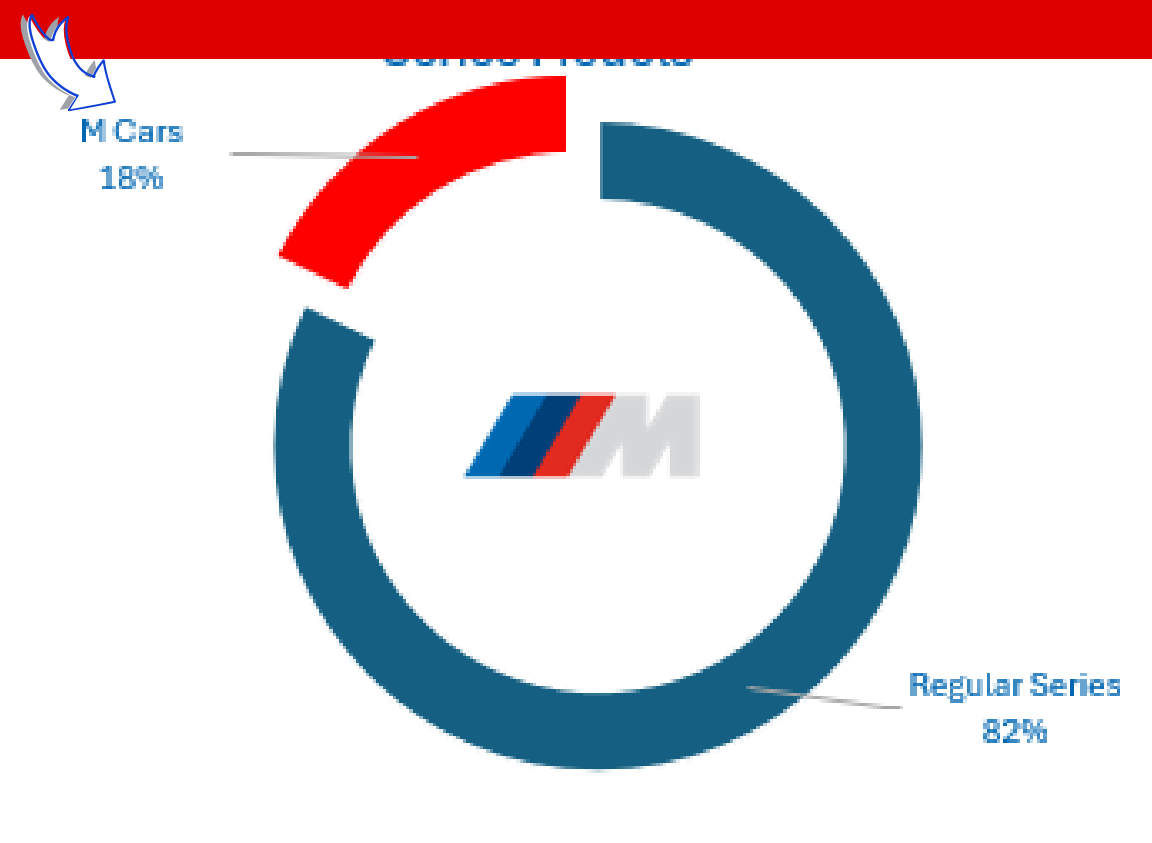
SLICERS AND CARDS



At the top of the dashboard, slicers allow filtering by body type, fuel type, and region, while a key performance indicator card highlights the total sales value of \$253 million in used BMW Cars

CHARTS:

1-USED CARS VS REGULAR SERIES



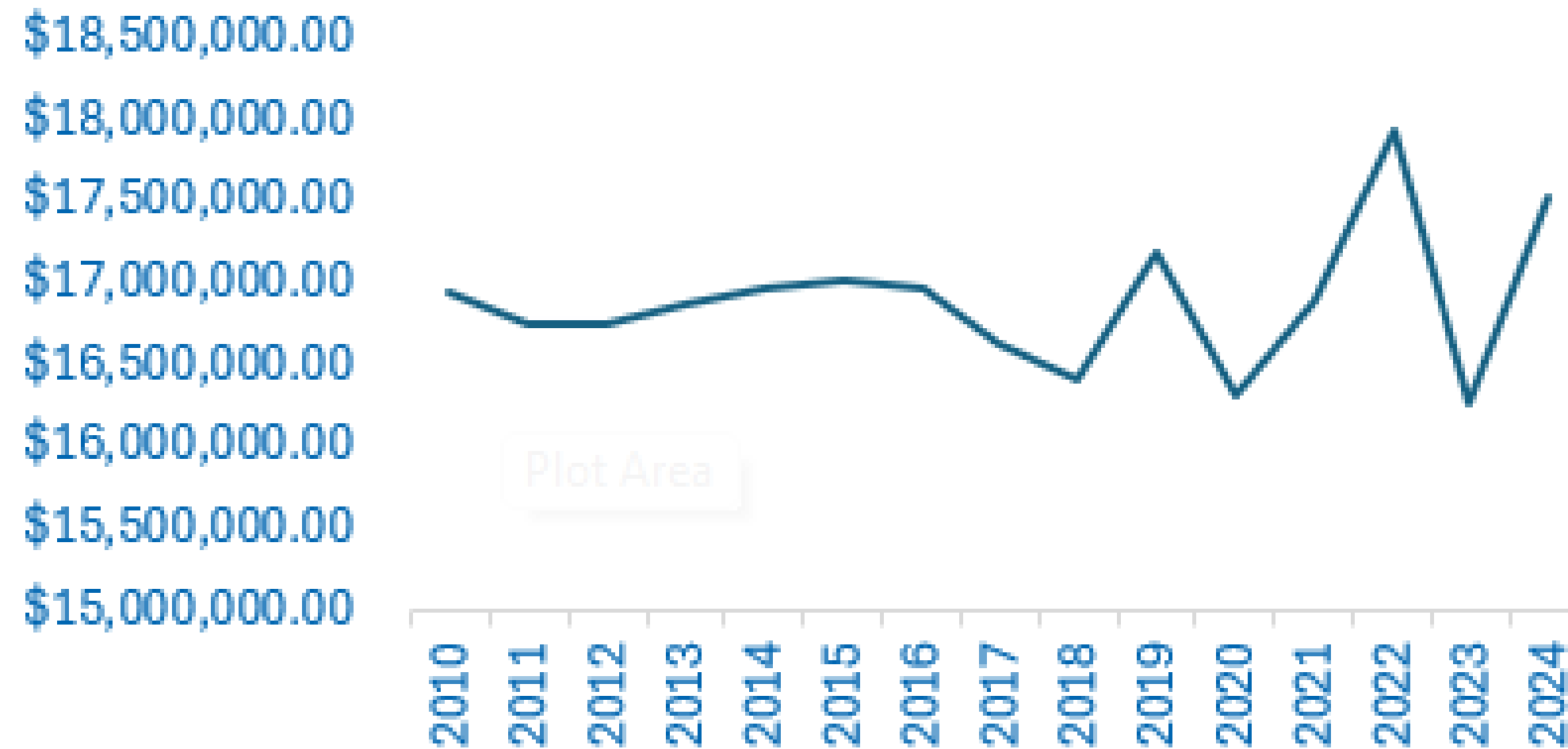
The comparison between BMW's high-performance M cars and the regular series models shows a clear imbalance in sales distribution. Regular Series models dominate the used market with 82% of total sales, reflecting their broader appeal to the used car customer segment due to affordability and practicality. In contrast, M cars, known for performance, premium features and high maintenance costs making up only 18% of sales, highlighting their incompatibility with the used cars market despite strong brand recognition.

2- TOP 3 REGIONS IN SALES



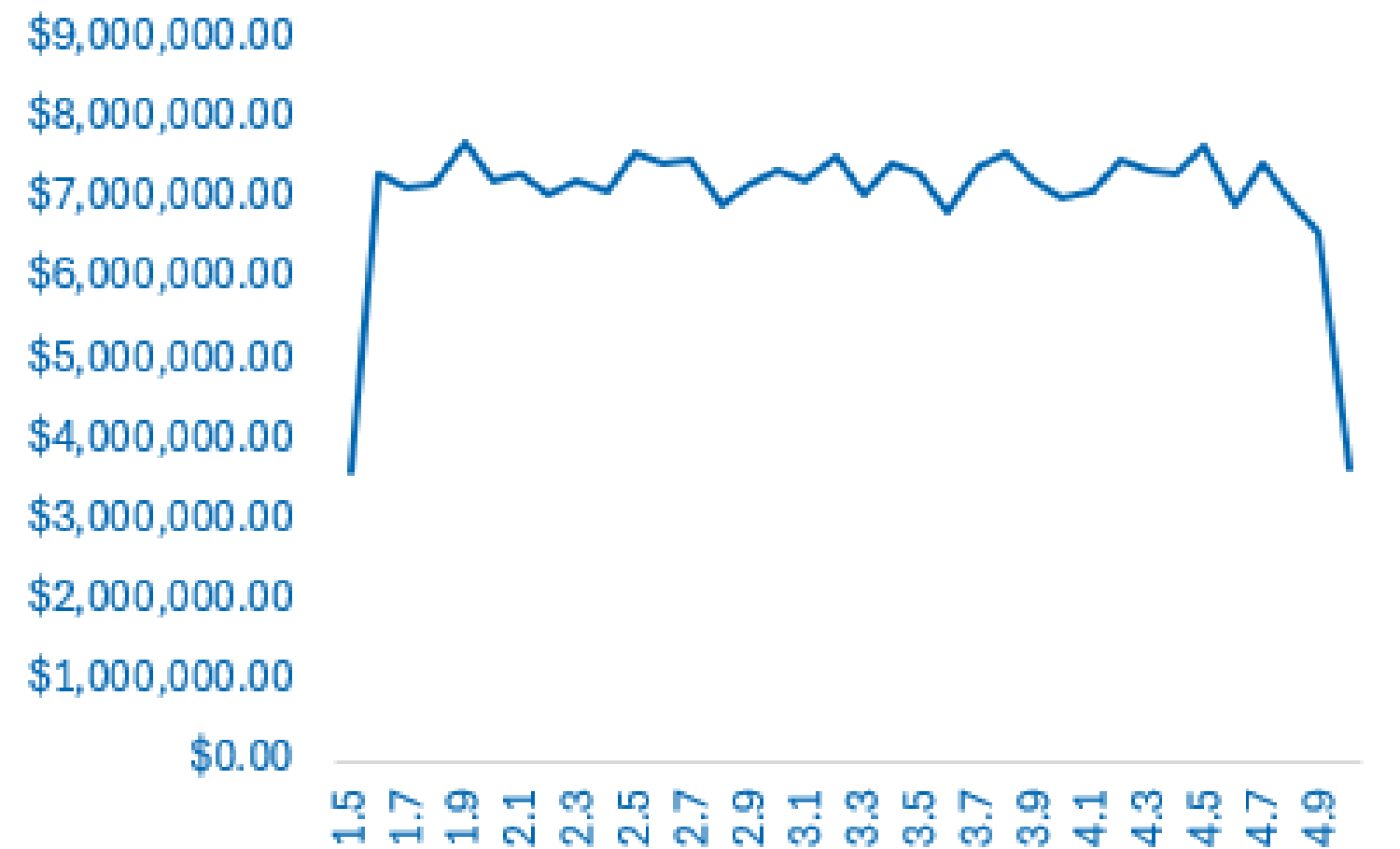
Regional analysis reveals that BMW's sales are concentrated in Asia (34%), North America (33%), and Europe (33%), which together account for all sales in the dataset. The near-even split suggests a strong global presence, with Asia slightly ahead, indicating growing demand in emerging and established Asian markets.

3. SALES BY MODEL YEAR



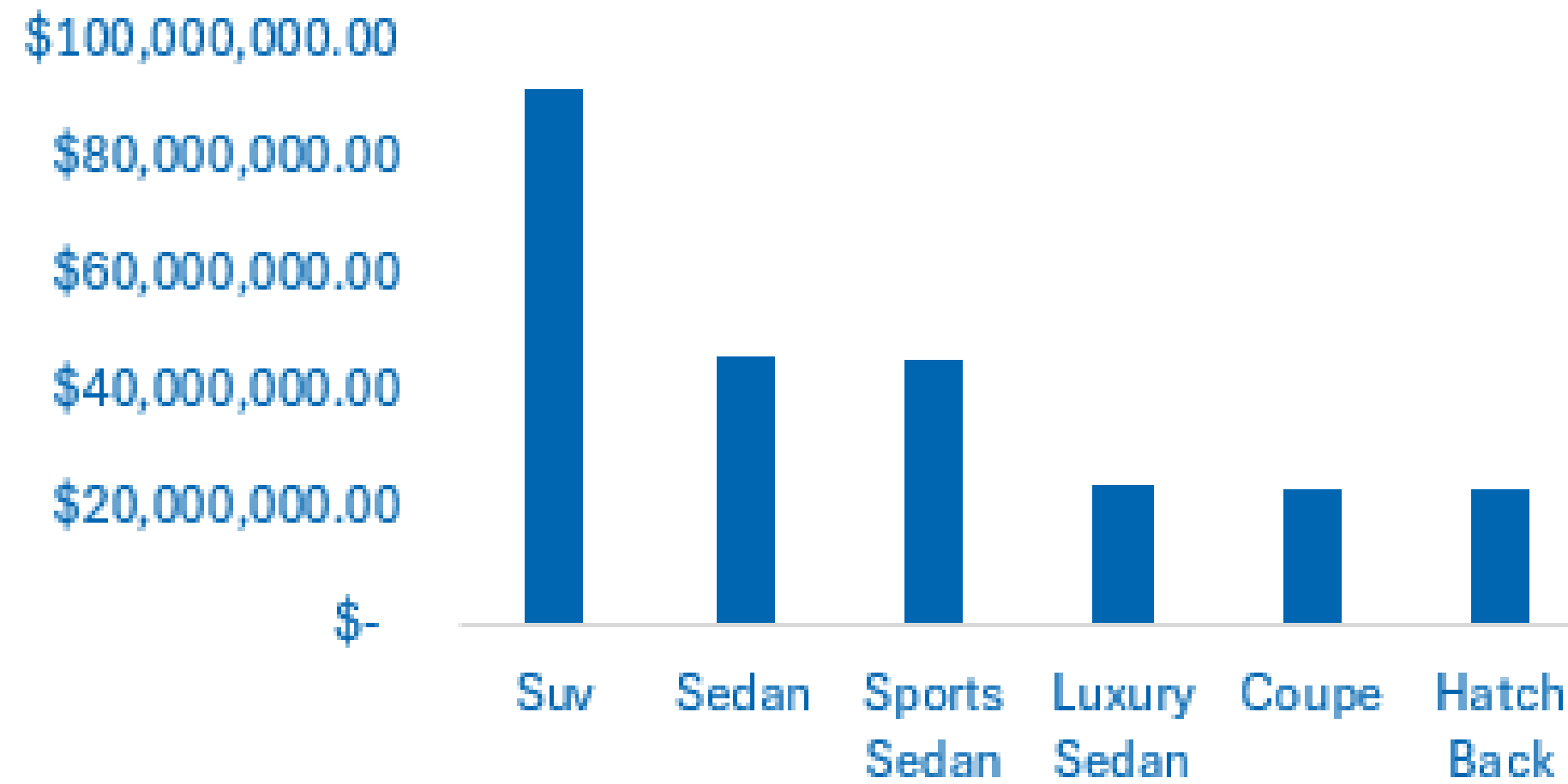
The sales trend by model year illustrates fluctuations in demand across time. Some years show clear low trend in sales volumes, possibly linked to the release of new models and strong economic conditions(ex 2016-2018) which engaged buying new models, while other years reflect increases that may be tied to weaker demand on new cars or economical/market factors. This visualization helps identify cycles in customer purchasing behavior.

4. SALES COMPARED TO ENGINE SIZE



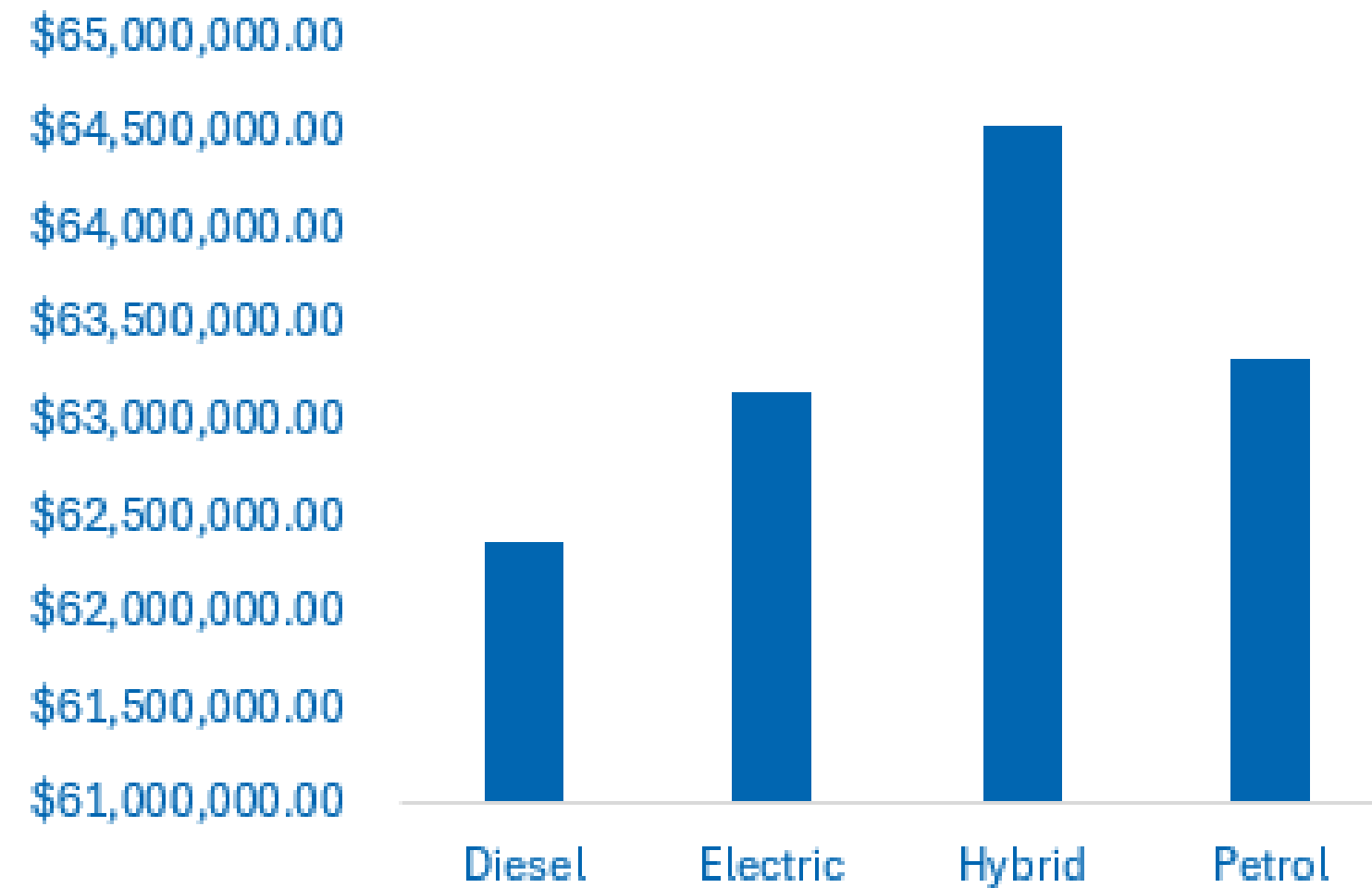
When examining sales by engine size, the data shows that sales are distributed relatively evenly across most engine categories except for low demand on highest and lowest Trims or engine Capacities Available , with no extreme dominance of a single size. This suggests BMW customers have diverse preferences, ranging from smaller engines that prioritize efficiency to larger engines that emphasize performance.

5. SALES BY CAR MODEL TYPE



The breakdown by car model type shows a strong preference for SUVs, which lead the market due to their versatility and popularity in both family and luxury segments. Sedans and Sports Sedans follow, indicating continued demand for traditional body styles. Meanwhile, Luxury Sedans, Coupes, and Hatchbacks contribute smaller shares, reflecting either niche positioning or lower demand compared to mainstream models.

6. SALES BY FUEL TYPE

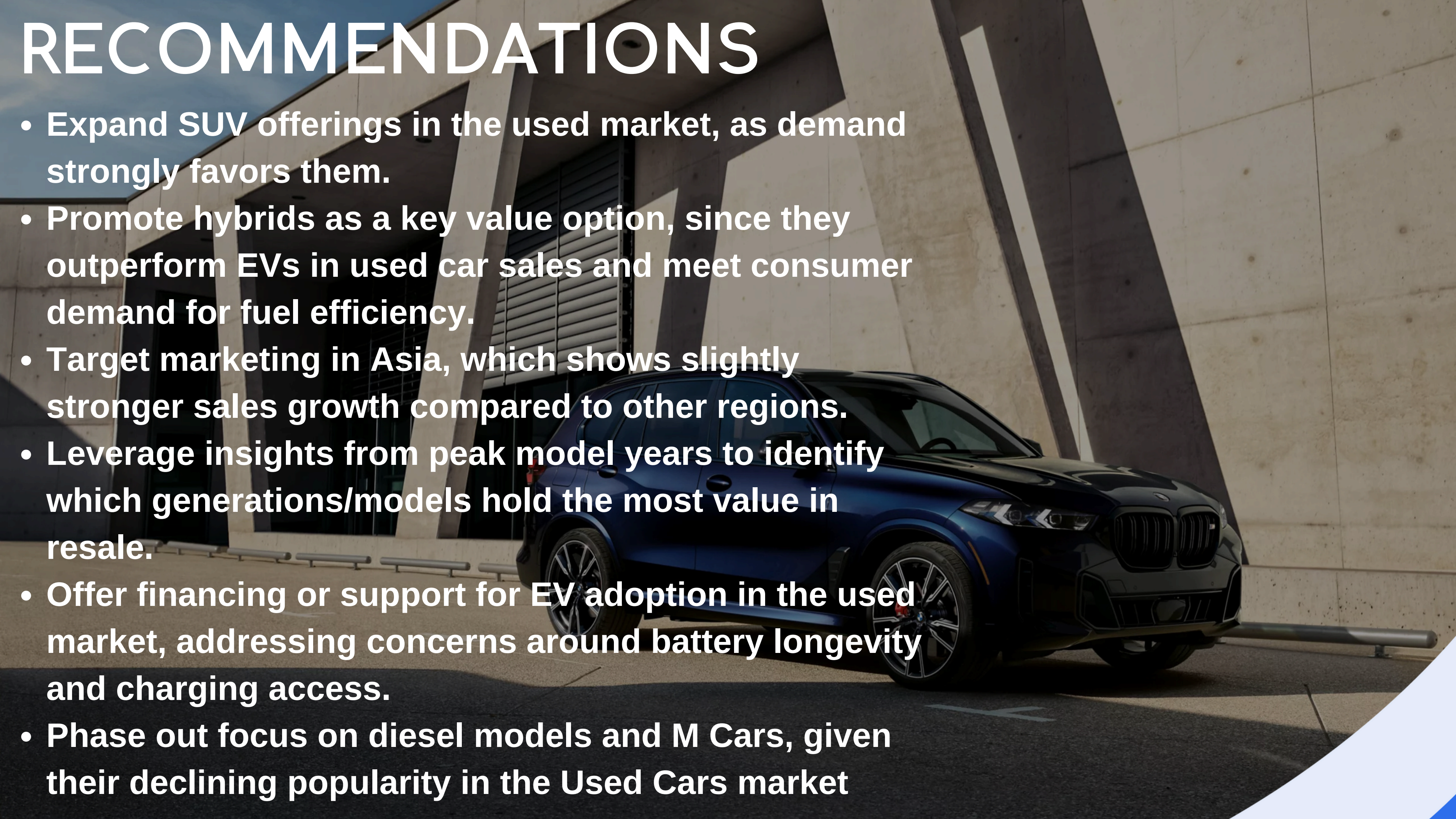


The fuel type analysis shows that Hybrids lead the used car market, outperforming both Petrol and Electric vehicles. This indicates that hybrid cars are gaining strong popularity in the secondhand market, likely due to their balance between fuel efficiency and affordability compared to fully electric models. Petrol cars still hold a solid share, reflecting their wide availability and established presence. Interestingly, Electric vehicles lag behind hybrids in the used market, which may be due to higher initial purchase prices, limited charging infrastructure in some regions, or concerns about battery life over time. Diesel cars represent the smallest portion, consistent with the global shift away from diesel technology..

IMPORTANT INSIGHTS

- Regular Series models dominate sales (82%), while M cars account for only 18%, showing their niche appeal.
- Asia leads slightly (34%), with North America (33%) and Europe (33%) almost equal, reflecting BMW's strong global presence.
- Sales vary significantly by model year, with peaks likely tied to new releases or favorable market conditions.
- Engine size demand is balanced, suggesting customers value both performance-oriented and efficiency-focused options in the used market.
- SUVs are the most popular body type, followed by Sedans and Sports Sedans, while Coupes, Hatchbacks, and Luxury Sedans trail behind.
- Hybrids are more popular than EVs in the used car market, signaling buyers prefer hybrids for cost-effectiveness and practicality, while diesel has the weakest demand.





RECOMMENDATIONS

- Expand SUV offerings in the used market, as demand strongly favors them.
- Promote hybrids as a key value option, since they outperform EVs in used car sales and meet consumer demand for fuel efficiency.
- Target marketing in Asia, which shows slightly stronger sales growth compared to other regions.
- Leverage insights from peak model years to identify which generations/models hold the most value in resale.
- Offer financing or support for EV adoption in the used market, addressing concerns around battery longevity and charging access.
- Phase out focus on diesel models and M Cars, given their declining popularity in the Used Cars market



THANK
YOU

